

Algebra II H - Unit 3 Solving Quadratics

Study Guide and Practice Guide

For extra practice - I suggest looking back at your notes and homework, and re-doing the examples for any topic you are struggling with.

Lesson #	Objectives:
Prior Knowledge	- I can factor quadratic trinomials when $a=1$ - I can factor quadratic trinomials when $a \neq 1$ - I can simplify radicals
03.1	- I can solve quadratic equations by graphing - I can solve quadratic equations by factoring - I can solve quadratics equations by square roots
03.2	- I can work with complex numbers. - Simplifying square roots with i - Adding and subtracting complex numbers - Multiplying complex numbers
03.3	- I can solve quadratic equation by completing the square - I can solve quadratic equations with complex solutions using - Complete The Square - Square Roots - Quadratic Formula - Given its roots, I can write a quadratic equation.
03.4	- I can find the discriminant of a quadratic equation and use that to explain the nature of its roots. - I can solve quadratic equations using the quadratic formula.
03.5	- I can solve systems of nonlinear equations - I can solve quadratic equations by graphing
03.6	- I can graph quadratic inequalities in two variables - I can solve quadratic inequalities in one variable
Unit 2 Review	- I can use quadratic equations to model real-world scenarios. - Given a scenario, write quadratic equation - Finding/interpreting vertex as max or min (CALCULATOR) - Finding/interpreting zeros (CALCULATOR) - Finding/interpreting y-intercept (CALCULATOR) - Finding/interpreting different values for a quadratic equation (CALCULATOR)
Unit 2 Review	- I can graph a quadratic function in standard and vertex form. - Given an equation in standard form, I can write it in vertex form and vice versa. - I can identify the vertex, axis of symmetry, domain, range, and increasing/decreasing intervals.
Unit 2 Review	- I can graph a quadratic function in intercept form. - Given an equation in intercept form, I can write it in standard and vertex forms. - Given a graph, I can write its corresponding equation.

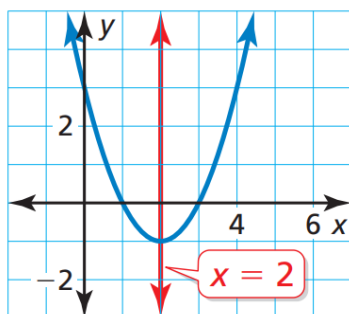
Write the expression as a complex number in standard form.

1. $(2 + i) - (3 + 4i) + 3i$

2. $(3 + 6i)(\frac{1}{3} - 2i)$

3. What are the powers of i circle? Draw a sketch and describe how you would utilize to find i^{105} ?

4. What does the graph tell you about the nature of its roots? **Unit 2 Review:** Given the graph write its corresponding equation (in all 3 forms)



Solve by factoring.

5. $3x^2 + 6x - 24 = 0$

6. $x^2 + 9x + 3 = -3x^2 + x$

Solve by square roots.

7. $3(x + 1)^2 = 27$

8. $-2x^2 - 98 = 0$

Solve by completing the square.

9. $x^2 - 12x + 4 = 0$

10. $4x^2 + 8x - 53 = 3$

11. $x^2 + 4x + 49 = 0$

12. $4x^2 + 72x + 388 = 0$

For each equation find the discriminant, indicate the meaning, and find the solutions using the quadratic formula.

13. $4x^2 - 8x + 5 = 0$

14. $3x^2 - 4x = 2x^2 + 2$

15. $2x^2 + x - 4 = 0$

16. Find a possible pair of integers values for a and c so that the quadratic equation has the given solution(s). Then write the equation.

a. $ax^2 - 4x + c = 0$; *two imaginary solutions*

b. $ax^2 + 6x + c = 0$; *two real solutions*

c. $ax^2 - 6x + c = 0$; *one real solutions*

17. Write an quadratic equation in standard form given the following answer from the quadratic formula

a. $x = \frac{-8 \pm \sqrt{-176}}{-10}$

b. $x = \frac{-9 \pm \sqrt{137}}{4}$

18. Solve the system by any method. Explain your choice of method.

a. $2x^2 - 2 = y$ & $-2x + 2 = y$

b. $x^2 - 6x + 13 = y$ & $-y = -2x + 3$

c. $x^2 + y^2 = 4$ & $-15x + 5 = 5y$

19. (GC) Solve $-3x^2 + 5x - 1 = 5x^2 - 8x - 3$ by graphing

20. Explain/Bullet how you solve a system of quadratic inequalities by graphing

21. Graph the inequality

a. $y > x^2 + 8x + 16$

b. $y \geq x^2 + 6x + 8$

c. $x^2 + y \leq 7x - 12$

22. Graph the system of quadratic inequalities

a. $x^2 - 4x + 8 > y$ & $-x^2 + 4x + 2 \leq y$

b. $2x^2 - x \geq y - 5$ & $0.5x^2 > y - 2x - 1$

c. $-3x^2 - 2x \leq y + 1$ & $-2x^2 + x - 5 > -y$

23. Solve the inequality

a. $3x^2 + 3x - 60 \geq 0$

b. $-x^2 - 10x < 21$

c. $3x^2 + 2 \leq 5x$

24. (GC) A rectangular parking lot must have a perimeter of 440 feet and an area of at least 8500 square feet. Describe the possible lengths of the parking lot.

25. (GC) Given $y = 2x^2 + 7x + 3$, find the x-value(s) where the y-value is 8.

26. (GC) Deion has 60 m of fencing to build a four-sided fence around a rectangular plot of land. The area of the land is 224 square meters.

a. Write a quadratic equation that expresses the area in terms of x, its length.

b. Solve for the dimensions (length and width) of the field.

27. (GC) A water balloon is catapulted into the air so that its height h , in meters, after t seconds

$$h = -4.9t^2 + 27t + 2.4.$$

a. How high is the balloon after 1 second?

b. What is the maximum height of the balloon?

c. When will the balloon burst as it hits the ground?

d. When is the balloon ascending? When is the balloon descending?

Unit 2 Review

28. Write the quadratic function (in all three forms) whose x-intercepts are at -4 and 0 and whose maximum value is 1.

Construct a quality graph of the quadratic function. Label the vertex and axis of symmetry. Identify domain, range, and increasing/decreasing intervals.

29. $y = 2x^2 - 4x + 3$

32. $y = -(x + 4)(x - 2)$

30. $y = -x^2 + 4$

31. $y = (x - 1)^2 + 2$

